

6(a)	The observation that most α – particle travel straight through the thin gold foil led to the conclusion that most of the gold foil and thus most of the atom is empty space. The observation that only a small proportion are deflected through an angle larger than 90° led to the conclusion that there is something massive and positively charged in the atom. Together with the first observation, this massive and positively charged region is very small in size relative to the atom. This region is called the nucleus.
6(b)	At minimum separation, the kinetic energy of the α – particle becomes electric potential energy.

	$1.23 \times 10^{-12} = \frac{(2e)(79e)}{4\pi\epsilon_0 d}$ $d = 2.96 \times 10^{-14} \text{ m}$
7(a)	When an electron absorbs the energy of a photon, it gains the necessary energy